

The Chinese University of Hong Kong
Division of Biomedical Engineering
Graduate Admission 2025 – 2026

Summary of Applicant's Information

Programme: ☒ Ph.D. in Biomedical Engineering ☐ M.Phil. in Biomedical Engineering																
CUHK Application No.: 25361191	HKPFS Application No. (if applicable): PF -															
HKPFS 1 st choice programme:	HKPFS 2nd choice programme:															
Intended Research Areas (Select at most two and number them by "1" and "2" in the order of your preference.) <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%; text-align: center;">1</td> <td style="width: 15%; border: 1px solid black;"> </td> <td style="width: 80%;">Biomaterials and Regenerative Medicine</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="border: 1px solid black;"> </td> <td>Biomolecular Engineering and Nano Medicine</td> </tr> <tr> <td style="text-align: center;">1</td> <td style="border: 1px solid black;"> </td> <td>Medical Imaging and Informatics</td> </tr> <tr> <td style="text-align: center;"> </td> <td style="border: 1px solid black;"> </td> <td>Medical Instrumentation and Biosensors</td> </tr> <tr> <td colspan="3">Others (please specify): _____</td> </tr> </table>		1		Biomaterials and Regenerative Medicine	2		Biomolecular Engineering and Nano Medicine	1		Medical Imaging and Informatics			Medical Instrumentation and Biosensors	Others (please specify): _____		
1		Biomaterials and Regenerative Medicine														
2		Biomolecular Engineering and Nano Medicine														
1		Medical Imaging and Informatics														
		Medical Instrumentation and Biosensors														
Others (please specify): _____																
Remarks (if any): _____																
Potential Supervisor (Please rank by "1", "2" and "3" in the order of your preference) 1) _____ Prof. Billy Wai-Lung Ng _____ 2) _____ Prof. WANG, Xiaogang _____ 3) _____ Prof. Owen H KO _____																

Part I - Personal Information			
Name in English: <u>ZHOU Xiangru</u>		Name in Chinese (if applicable): <u>周相如</u>	
Date of Birth: 31 (day)/ 07 (month)/ 1990 (year)	Sex: ☒ Male ☐ Female		
Contact Address: <u>RM 5D, 2nd floor, 5 Humbert St, Kowloon, Hong Kong.</u>			
Email: <u>xiangru.zhou@connect.polyu.hk</u>	Contact Tel.: <u>+86 18502091962</u>	Mobile Phone No.: <u>+852 64820616</u>	

Part II – Education			
English Test Scores: TOEFL: IELTS <u>6.5</u> CET: _____ GRE (Verbal): _____ Other: _____			
1) Bachelor Degree		2) Master Degree	
Name of the University: (If it is a university in Mainland China, give the Chinese name) <u>西安理工大学</u>		Name of the University: (If it is a university in Mainland China, give the Chinese name) <u>The Hong Kong Polytechnic University</u>	
Major: <u>Microelectronics</u>		Major: <u>Microelectronics Technology & Materials</u>	
Year of Graduation: <u>2013</u>		Year of Graduation: <u>2025</u>	
GPA: _____ / 4.0	GPA: _____ / 4.0		_____ / 100
Rank in Class: _____ out of _____ students		Rank in Class: _____ out of _____ students	
Top _____ %		Top _____ %	

Part III –Publication Record	
Title of Bachelor Thesis: _____	
Title of Master Thesis: _____	
No. of papers published in international journals: _____	
No. of papers published in local journals: _____	
No. of papers published in international conferences: _____	
No. of papers published in regional conferences: _____	
No. of other research outputs (e.g. patent, etc.): <u>12 Patents</u>	

Part IV – Awards	
Give 3 most important academic awards you have obtained	
1) _____	
2) _____	
3) _____	

Part V	
Through which channel, you know this admission exercise:	
☒ our web ☐ search engine (Google, Yahoo, etc) ☐ peers ☐ others (please specify: _____)	

Xiangru Zhou

Email: xiangru.zhou@connect.polyu.hk | Telephone: (+86) 18502091962, (852) 64820616

CAREER PROFILE

10+ years experience in image processing. I have a proven track record of innovation and excellence. My expertise spans the image processing industry, especially in programming and computer vision, where I have deep knowledge of process flows and a history of leading practical application projects to success. I've applied for over **10 patents** and am driven by the promise of technologies like Brain-Computer Interfaces (BCI) and medical image processing to improve lives.

EDUCATION

The Hong Kong Polytechnic University

MSc Microelectronics Technology & Materials

Hong Kong

Sep. 2024 – Present

Xi'an University of Technology

Bachelor of Microelectronics

Xi'an, China

Sep. 2009 – Jun. 2012

• **GPA: 3.64/5.0**

• **Average Score:** 81.18/100

• **Honours:** 2nd Prize 8th Xi'an High-tech "Challenge Cup" Shaanxi Province University Student Extracurricular Academic Science and Technology Works Competition; 1st and 3rd Prize University 19th Extracurricular Academic Science and Technology Works Competition; 1st Prize University 18th "LiAo Cup"; University Innovation Achievement 2nd Prize, University 3rd Prize Scholarship

PROFESSIONAL EXPERIENCE

SMARTMORE CORPORATION LIMITED

Software Engineer

Shenzhen, China

May 2020 – Sep. 2024

AI-Driven Robotic Arm Project

Leader

May 2024 – Sep. 2024

• Branch 1: Traditional Hand-Eye Calibration + Image Recognition

- Implemented 3D scenarios: arbitrary object recognition and grasping, sketching based on verbal commands, and interactive stacking toys with users.
- Integrated an Intel Realsense camera, a structured light 3D camera, a 7-axis robotic arm with a gripper, two servers for large-model deployment, and a PC for control and ASR-based voice input.
- Deployed conversational and vision models to enable seamless interaction between visual, auditory, and motor systems.
- Achieved high precision in object localization and vision-guided robotic operations.

• Branch 2: Transformer-Based Action Prediction

- Utilized a Transformer model to predict robotic arm actions by joint data, CNN features, and task descriptions.
- Enabled real-time responses without hand-eye calibration, adapting to dynamic environments.
- Tackled computational challenges and dataset requirements for effective training and deployment.

Wafer ID Reader Wafer OCR Code Reader Integrated Machine Project

Leader

Oct. 2021 – Sep. 2024

- The first deep learning-based OCR wafer character recognition tool in the semiconductor industry. Convenient use and no need to adjust the recognition parameters compare to its competing products.
- Collaborated with the algorithm team to devise strategies, conduct experiments, and select optimal algorithmic solutions.
- Modified software and SDK interfaces and functionalities based on the product manager's research and user requirements.
- Implemented end-to-end closed-loop process from requirement analysis to deployment, including evaluation, experimentation, algorithm/software/SDK development, testing, and deployment.
- Same recognition rate level as that of industry leader Cognex products in normal cases, partially exceeds its recognition rate (100% versus 99.5%) in the case of fixed format recognition.

Defect Detection System for SONY Labels

Second-hand Collaborator

Feb. 2023 – May 2023

- Conducted Proof of Concept (POC) using client samples to validate system performance.
- By combining the template image and the image to be checked together to form a two-channel image, defects are marked and trained on this basis, the accuracy has been improved by 2%.

BGI Genomics Reagent Bottle Testing

Collaborator

Oct. 2022 – Feb. 2023

- Designed algorithmic solutions and implemented the AI model into software for real-time monitoring of reagent bottle caps, ensuring seal integrity and smoothness during the manufacturing process.
- Conducted Optical Character Recognition (OCR) for character detection on reagent bottle bodies, effectively identifying characters on various colored backgrounds of medication and detecting printing defects.

Specific String OCR System for Arbitrary Orientations On Tags

Designer

Jun. 2021 – Feb. 2022

- Designed the composition of models responsible for the task, leveraging the existing MobileNet training framework in PyTorch. Trained four models individually: label localization, orientation determination, character localization, and recognition.

- Developed a software development kit (SDK) in Visual Studio to facilitate integration and usage of the OCR system.
- Designed and implemented a user-friendly interface using the QT framework to enhance the usability and accessibility of the OCR system.

Character Recognition System for Apple Watch Bands Using Laser Engraving Technology

Designer

May. 2020 – May 2021

- Trained character localization and classification models on a four-card 2080Ti server running Ubuntu, utilizing the MobileNet architecture.
- Employed the ONNX Runtime (ORT) for model deployment, integrating a C++ SDK for inference on the server and developing a user-friendly software interface.
- Collaborated with colleagues to align and optimize the SDK, including pre-processing and post-processing workflows.
- Precision and recall were above 99.7% respectively.

Defect Detection System for Apple Watch Bezels Using 3D Imaging Technology

Designer

Jun. 2020 – Jun. 2020

- Implemented the conversion of 3D point cloud data into 2D heatmaps for analysis purposes.
- Designed and programmed the user interface for the software application.

SHENZHEN PHDI CORPORATION LIMITED

Co-founder, CTO

Shenzhen, China

Jan. 2017 - May 2020

Android Software for A Car-Mounted Mobile Phone Controller

Designer

Jan. 2017 – May 2020

- Managed PCB production designed Bluetooth chip-related programs and conducted 3D design and printing for the product's appearance.
- Enabled rapid switching between navigation, music, and WeChat applications in a car environment.
- Implemented automatic sending and receiving of WeChat messages.
- Integrated features for one-touch activation of voice navigation and song search.
- Designed and developed a comprehensive product including a mobile application and a Bluetooth controller.

Face Recognition and Tracking Software on The Android Platform

Designer

Jan. 2017 – May 2017

- Utilized OpenCV, Caffe, Idlib, and VGG net deep learning neural network for face localization and gender recognition.
- Integrated BP neural network for gender recognition within the face detection algorithm.
- Applied the developed algorithm in systems for estimating the attention of different gender groups towards billboards and for access control systems, resulting in the acquisition of a patent.
- Adapted the algorithm for integration with security cameras to track familiar/strange faces indoors, aiding in the detection of potential security threats.
- Successfully delivered the developed solution to a renowned domestic security camera company.

OPT MACHINE VISION TECH CO., LTD.

Image processing algorithm engineer

Guangdong, China

Apr. 2014 – Jan. 2017

One-Dimensional Barcode Localization and Recognition Algorithm

Designer

Apr. 2014 – Jan. 2017

- Achieved automatic localization of barcodes in approximately 10ms on a PC equipped with an i3 processor, capable of identifying barcodes in complex images of up to 2 million pixels.

Two-Dimensional Barcode Localization and Recognition Algorithm

Designer

Jan. 2015 – Jan. 2017

- Implemented functionality to locate and recognize QRCode, DataMatrix, and other two-dimensional barcodes.
- Utilized Google's open-source libraries ZXing/Zbar and made necessary modifications for enhanced performance

Rapid Edge And Circle Detection Algorithm

Designer

Jan. 2015 – Jan. 2017

- Utilized non-Hough fitting method to swiftly fit specified parameters of lines and circles in designated directions, implementing the algorithm to effectively recognize lines and circles in blurry edge images of up to 2 million pixels.
- Achieved real-time identification within 10ms on a PC with an i3 processor and 8GB of memory.

ADDITIONAL INFORMATION

-
- **Language:** Cantonese, Mandarin, English (Fluent; IELTS: 6.5), Japanese (N5)
 - **Programming Language:** C, C++, Python, Java, Java Script, HTML, CSS, PHP
 - **Development Tools & Libraries:** QT, Visual Studio, gdb/pdb, Git, Docker, Anaconda & OpenCV, PyTorch, Halcon
 - **Other Skills:** Keil (STM32), Altium Designer (PCB), Rhino (3D printing), Dreamweaver (Website)
 - **Books:** Computer Vision: Algorithms and Applications, Design Patterns: Elements of Reusable Object-Oriented Software
 - **Patents:** CN202311136494.1; CN202310973415.6; CN202310287581.0; CN202310051665.4; CN202211602503.7; CN202211474648.3; CN202211357774.0; CN202210994137.8; CN202230264395.1; CN201911330078.9; CN201110120104; CN201110308514.X

香港居民身份證

HONG KONG IDENTITY CARD

周相如

ZHOU, Xiangru

F601647

0719 4161 1172

出生日期 Date of Birth

31-07-1990 男 M

CX

簽發日期 Date of Issue

(09-24)

19-09-24 F601647(9)









打印日期:2024年6月4日

[illegible]

Date: Jun. 4, 2024

西安理工大学本科课程考核成绩与绩点换算关系

西安理工大学本科课程考核成绩、课程绩点的换算关系如下表：

百分制	95-100	85-94	75-84	65-74	60-64	< 60
对应课程绩点	5.0	4.0-4.9	3.0-3.9	2.0-2.9	1.5-1.9	0.0
五级制(百分制) 绩点	优(90) 4.5		良(80) 3.5	中(70) 2.5	及格(60) 1.5	不及格 0.0
二级制(百分制) 绩点	合格(75) 3.0					不合格 0.0

考核成绩的记载一般采用百分制,实践性课程可采用五级记分制(即优、良、中、及格、不及格),也可采用二级记分制(合格、不合格)。课程考核成绩及格以上计为合格,即取得该门课程的学分,不合格没有学分(以“0”学分记)。

学分绩点的计算方法具体为:

课程学分绩点=课程绩点×课程学分

平均学分绩点=课程学分绩点总和÷课程学分总和。

The following table is the conversion between course grades and course grade points at Xi'an University of Technology.

Scores	95-100	85-94	75-84	65-74	60-64	< 60
Grade Point	5.0	4.0-4.9	3.0-3.9	2.0-2.9	1.5-1.9	0.0
Five-Level Basis Grade Point	Excellent (90) 4.5		Good (80) 3.5	Average (70) 2.5	Pass (60) 1.5	Fail 0.0
Two-Level Basis Grade Point	Pass(75) 3.0					Fail 0.0

Grades are usually recorded on a percentage basis, while practical courses may use a five-level grading system (i.e., excellent, good, average, pass, fail), and a small number of courses may use a two-level grading system (i.e., pass, fail). The passing grade for a course is considered qualified, which earns credit for that course, while failing earns no credit (recorded as "0" credit).

The calculation of grade point average (GPA) is as follows:

Course credit point = course grade point * course credits

Average GPA = total course credit points / total course credits.



Xi'an University of Technology



学士学位证书

周相如，男，1990年7月31日生。在西安理工大学
微电子学
专业完成了本科学习计划，业已
毕业，经审核符合《中华人民共和国学位条例》的规定，授予工学
学士学位。

西安理工大学

校长

学位评定委员会主席

证书编号：1070042013001855

二〇一三年七月三日

(普通高等教育本科毕业生)



普通高等专科学校

毕业证书



学生 周相如 性别 男，一九九〇年七月三十一日生；于二〇〇九年

九月至二〇一三年七月在本校 微电子学

专业 四年制 本科 学习，修完教学计划规定的全部课程，成绩合格，准予毕业。



校 名：

西安理工大学

校（院）长：



证书编号： 107001201305701815

二〇一三年七月三日

Test Report Form

ACADEMIC

NOTE Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules.
GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes.
It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test.

Centre Number

CN002

Date

10/MAR/2024

Candidate Number

137676

Candidate Details

Family Name

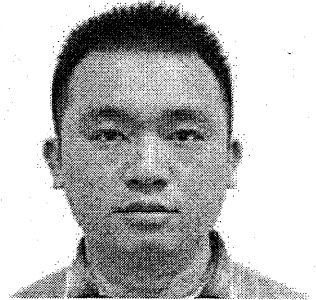
ZHOU

First Name

XIANGRU

Candidate ID

450881199007318019



Date of Birth

31/07/1990

Sex (M/F)

M

Scheme Code

Private Candidate

Country or Region
of Origin

CHINA (PEOPLE'S REPUBLIC OF)

Country of
Nationality

First Language

CHINESE

Test Results

Listening

6.0

Reading

6.5

Writing

6.0

Speaking

7.0

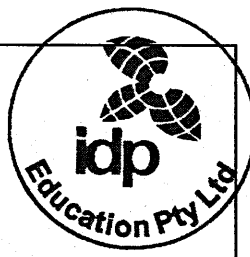
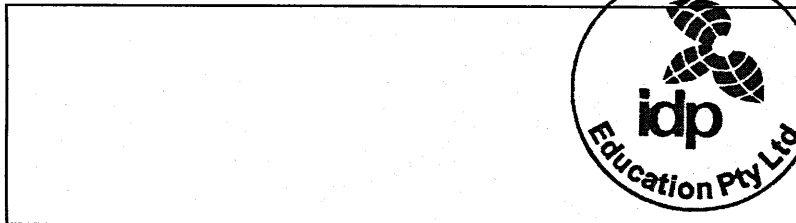
Overall
Band
Score

6.5

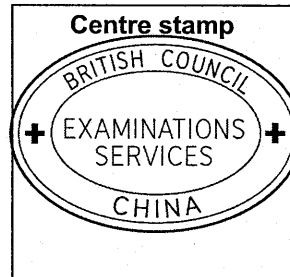
CEFR
Level

B2

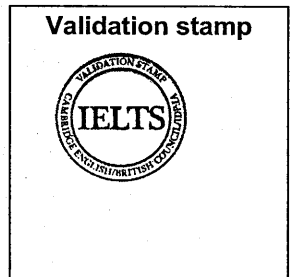
Administrator Comments



Centre stamp



Validation stamp



Administrator's
Signature

Date

13/03/2024

Test Report Form
Number

23CN137676ZH0X002A



Cambridge Assessment
English

Advancing Brain-Computer Interface (BCI) Technology

As a Master's student in Microelectronics Technology and Materials at The Hong Kong Polytechnic University, I have cultivated a multidisciplinary skill set spanning image processing, AI-based image recognition, and electronics engineering. My decade-long professional experience includes roles such as Algorithm Engineer and R&D Lead in notable organizations like OPT and SmartMore Technology. These experiences equipped me with expertise in software development (C++ and Python), model optimization, data preprocessing, and system integration. Additionally, I hold over ten invention patents and have a strong foundation in designing and prototyping Bluetooth communication devices and robotic systems.

Brain-Computer Interface (BCI) technology has been a long-standing passion of mine, representing a confluence of engineering, materials science, and neuroscience. This interest has guided my academic and professional pursuits, inspiring my transition from software and AI applications to semiconductor materials and microphysics. By addressing the challenges of electrode precision and signal acquisition, I aim to advance the field of BCI, which I believe holds transformative potential in healthcare, rehabilitation, and human-computer interaction.

To align my expertise with the demands of BCI research, I have focused on strengthening my interdisciplinary knowledge. My Master's research has enabled me to delve into nanomaterials and microphysics, providing essential insights into the interaction between biological signals and electrode materials. This foundation has prepared me to investigate advanced signal acquisition techniques, refine AI-driven signal recognition models, and explore the integration of BCI with robotics—particularly in developing brain-controlled prosthetics.

If admitted to CUHK's PhD program, I aim to contribute to its vibrant research ecosystem through the following objectives:

1. **Data Acquisition and Signal Analysis:** Collect and preprocess large-scale neural datasets, ensuring data quality through advanced cleaning and classification methods.
2. **AI Model Development:** Design robust neural signal recognition models to improve accuracy and stability, leveraging CUHK's strengths in AI research.
3. **Real-World Applications:** Investigate practical implementations of BCI, particularly in robotics, to create innovative rehabilitation tools.
4. **Interdisciplinary Collaboration:** Work closely with neuroscientists, material scientists, and medical professionals to advance electrode technologies and enhance BCI precision.

I am acutely aware of the challenges in this field, including the low initial accuracy of signal recognition models and the variability in device performance across environments. My professional background has taught me the importance of addressing such challenges through data quality enhancement, precision engineering,

and cross-disciplinary teamwork. By gaining deeper insights into neuroscience and medical science, I am committed to bridging the gap between technology and healthcare solutions.

CUHK's pioneering research in neuroscience and biomedical engineering aligns perfectly with my aspirations. Its collaborative and innovative research environment provides an ideal platform to advance BCI technology. I look forward to contributing to CUHK's research community and making a meaningful impact in this transformative field.



【奖学金】

Scholarship

证书

Certificate

Certificate

祖国 荣誉 责任

周相如

荣获 2011 — 2012 学年

三等

奖学金

*This is a certificate conferred
to Zhou Xiangru
for 2011 - 2012 academic year
Third Prize
Scholarship.
2012.11*

西安理工大学
XI'AN UNIVERSITY OF TECHNOLOGY